



秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

檢測報告

Test Report



Chang Bing Show Chwan





瑞德生物科技有限公司 MASTER LABORATORY CO.,LTD.

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瑞德生物科技有限公司
MASTER LABORATORY CO., LTD.

In Vitro Cytotoxicity Test

Master Laboratory Co., Ltd. Genetic Toxicology Testing Laboratory

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RELANO WOUND SPRAY
***In Vitro* Cytotoxicity Test**
STUDY REPORT

Sponsor : Nanoplus Life Biomedical Technology Co., Ltd.

Testing Institution : Master Laboratory Co., Ltd.

June 2023



FINAL REPORT

Report No.: 23S186CE-01-R01

Report No.: 23S186CE-01-R01

Experimental starting date: 06.12.2023

Test article registration date: 06.16.2023

Test article extraction date: 06.19.2023

Experimental completion date: 06.21.2023

Study Announcement

1. The study report is valid for the test article used only.
2. The study report could not be recopied or extracted only if the permission from Master Laboratory Co., Ltd.
3. The study report is invalid without the endorsement of Master Laboratory Co., Ltd.



SIGNATURE OF STUDY PERSONNEL

Study Director

Jyun Cheng Liu
Jyun Cheng Liu

06.30.2023
Date

Facility Management

Alan Hsieh
Alan Hsieh

06.30.2023
Date

**GLP COMPLIANCE STATEMENT**

The study met with the technical requirements of the protocol, and the applicable guidance, which included the Good Laboratory Practice for Non-Clinical Laboratory Studies (FDA, 21 CFR, Part 58, 2019), OECD Principles of Good Laboratory Practice (No. 1, 1997) and Good Laboratory Practice for Non-Clinical Laboratory Studies (Food and Drug Administration, R.O.C., 2019), except the parts of test article identification and related analyses (21 CFR §58.105, §58.113, FDA; 6.2.2-6.2.5, OECD GLP No.1, 6.2.2-6.2.5, TFDA GLP). To the best of our knowledge, there were no deviations from the approved study plan and no adverse problem that would affect the integrity of this study or the interpretation of the study result.

Study Director

Jyun Cheng Liu
Jyun Cheng Liu

06.30.2023
Date

**QUALITY ASSURANCE STATEMENT**

The Quality Assurance personnel had inspected the conduct of different phases of the study according to a predetermined study schedule. To the best of our knowledge, there were no deviations from the study plan and standard operating procedures that would affect the integrity of this study. This report has been audited by the QA personnel in accordance with the appropriate standard operating procedures of Master Laboratory Co., Ltd. described the methods and procedures used in the study, and the reported results accurately reflect the raw data generated during this study.

Listed below are the phases in this study that were audited by the QA personnel and the dates the audits were performed and findings reported to facility management (FM).

Inspection record:

Date of inspection	Phase	Date Reported to SD	Date Reported to FM
06.12.2023	Study protocol, Personnel quality and test article. (Study base)	06.12.2023	06.12.2023
06.26.2023	Study system, study report, raw data, and final report. (Study base)	06.26.2023	06.26.2023

Quality Assurance unit in charge


Ying Chun Chen


Date



Contract Research Organization and Sponsor Information

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SUMMARY

The present study was to evaluate the in vitro cytotoxicity of extract from the test article "RELANO WOUND SPRAY" after 24h treatment. The testing was performed in compliance with ISO 10993-5:2009 and EN ISO 10993-5:2009. The cell morphology of mouse fibroblast cell line (L929) and the confluency of monolayer were scored under the microscope for qualitative determination. The MTT assay was performed in order to check the cytotoxicity of the test article extract quantitatively. Both qualitative and quantitative results showed that no cytotoxicity of the test article extract was observed in this study.



INTRODUCTION

Evaluation of the *in vitro* cytotoxicity of a medical device was the initial step of a biocompatibility study. It was usually performed using immortalized cell lines, for example, L929. The aim of this study was to investigate the cytotoxic effect of the test article extract. The cytotoxicity was determined qualitatively and quantitatively.

TEST ARTICLE

A. Test article identification (Based on information provided by sponsor)

Test article: RELANO WOUND SPRAY

Test article Lot No. 51223050003

Sterilization condition: Not Sterilized

Expiry day: 05.26.2025

B. Characteristics of Test article

Physical description	Liquid
Major composition	Nano Ion Water, Sodium Carboxymethylcellulose
Test article as received or treated	The test article was extracted as received without pretreatment.

C. Extraction condition of test article

According to ISO 10993-12:2021, the extraction ratio of test article weight/culture medium volume was approximately 0.2 g/mL.



MATERIALS AND METHODS

A. Chemicals and Materials

1. Cell system

L929 cells were obtained from Food Industry Research and Development Institute (BCRC RM60091), referred to as L929 mouse fibroblasts.

2. Medium and reagents

- a. MEM/Alpha modification (Hyclone, SH30568.01, AH30060188)
- b. Horse Serum (GIBCO, 16050-122, 2582144)
- c. Penicillin-Streptomycin solution (GIBCO, 15140-122, 2441868)
- d. L-glutamine (GIBCO, 25030-081, 2492879)
- e. MEM-non essential amino acid (Hyclone, SH30238.01, AH30708708)
- f. DPBS (Hyclone, SH30256.02, AH30708720)
- g. MTT cell proliferation/viability assay kit (R&D systems, 4890-25-K, Reagent LOT: P314211, Detergent LOT: P314214)
- h. Neutral red (SIGMA-ALDRICH, N2889, RNBK5997)
- i. Trypsin 0.25% (1X) solution (Hyclone, SH30042.01, J210038)
- j. Trypan Blue stain 0.4% (invitrogen, T10282, 2433858)
- k. Dimethyl sulfoxide (SIGMA-ALDRICH, D2650, RNBK5742)

3. Controls

- a. Blank(B): culture medium (MEM/ alpha modification inclusive of 10% (v/v) horse serum, 1% (v/v) penicillin-streptomycin solution, 1% (v/v) L-glutamine and 1% (v/v) non-essential amino acid)
- b. Negative control (NC): high density polyethylene (HDPE)
- c. Positive control (PC): culture medium inclusive of 10% (v/v) DMSO



B. Methods

1. Cell culture

The cryopreserved L929 cells were thawed, and cultured in MEM/alpha modification medium at $37\pm 1^{\circ}\text{C}$ in 5% CO_2 atmosphere. Up to 80% cell growth was observed by microscope, subculture until obtaining passage number 2 prior to use. The culture medium should be replaced twice a week.

2. Sample preparation

2.1. Test Sample:

According to ISO 10993-12:2021, the extraction ratio of test article weight/culture medium volume shall be 0.2 g/mL, extract at a rotation speed of 100 rpm under $37\pm 1^{\circ}\text{C}$ for $24\pm 2\text{h}$. After extraction, the extract solution was used immediately; the appearance of test article extract was clear, particulate-free and no color changed.

Test article group 1: Test article extract, original extract

Test article group 2: 50 % extract of the test article

2.2. Positive control (PC):

Culture medium with 10% (v/v) DMSO, and incubated at a rotation speed of 100 rpm under $37\pm 1^{\circ}\text{C}$ for $24\pm 2\text{h}$.

2.3. Negative control (NC):

Based on the ISO 10993-12:2021, the extraction ratio was 0.2 g/mL, took 1.0g HDPE, immersed it in 5.0 mL culture medium, and incubated at a rotation speed of 100 rpm under $37\pm 1^{\circ}\text{C}$ for $24\pm 2\text{h}$.

2.4. Blank (B):

Took 5.0 mL of culture medium and then incubated at a rotation speed of 100 rpm under $37\pm 1^{\circ}\text{C}$ for $24\pm 2\text{h}$.



3. Qualitative and quantitative (MTT assay) determination

a. Qualitative analysis

5×10^4 of L929 mouse fibroblast cells were seeded in 24-well culture plates and then incubated at $37 \pm 1^\circ\text{C}$ in 5% CO_2 atmosphere for $24 \pm 2\text{h}$. After cell attachment in a subconfluent monolayer, the original culture medium was removed and replenished 0.5mL of B, NC, PC and Test article group 1 in each of culture well. Each treatment was carried out in triplicate. Then the test plates were subsequently incubated at $37 \pm 1^\circ\text{C}$ in 5% CO_2 atmosphere for $24 \pm 2\text{h}$. The plate was incubated for 24h. After 24h, the cells of each well were stained with the neutral red solution and scored by the change of cell morphology and viability under inverted microscope in accordance with criteria of appendix 1.

b. Quantitative analysis (MTT assay)

L929 mouse fibroblast cells were seeded in 96-well culture plates with the number of 1×10^4 cells, and then incubated at $37 \pm 1^\circ\text{C}$ in 5% CO_2 atmosphere for $24 \pm 2\text{h}$. After 24h, the original culture medium was removed and replenished 0.1mL of B, NC, PC, Test article group 1 and Test article group 2 in each of culture well. Each treatment was carried out in triplicate. Then the test plates were subsequently incubated at $37 \pm 1^\circ\text{C}$ in 5% CO_2 atmosphere for $24 \pm 2\text{h}$. At the end of the incubation period, the B, NC, PC, Test article group 1, Test article group 2 was removed and replenished 0.1mL of fresh culture medium. Then 10 μL MTT reagent (kit component, 4890-25-01) was added into each well. The reaction was performed at 37°C in 5% CO_2 atmosphere for 2-3h and kept from light. 0.1mL of detergent reagent (kit component, 4890-25-02) was added into each well and incubated in dark for 2-3h before measuring the absorbance. The absorbance of test samples was measured at 570 nm (reference wavelength: 630 nm) by a microplate reader (ELx800,



BioTek). The absorbance of test samples was represented as mean $\pm$ standard deviation (SD).

Compared to the blank (as 100% viability), the percentage of viability and mortality was calculated by the following formula:

$$\text{Viability (\%)} = \frac{\text{OD}_{570e}}{\text{OD}_{570b}} \times 100\%$$

$$\text{Mortality (\%)} = 100 - \text{Viability (\%)}$$

Where

OD_{570e} is the mean value of the measured absorbance of the test samples;

OD_{570b} is the mean value of the measured absorbance of the blank (B).

If viability is reduced to $< 70\%$ of the blank, indicating the test article has a cytotoxic potential.

C. Criteria for a valid test

1. A test meets the acceptance criteria if the mean OD570 of blanks is ≥ 0.2 .
2. The left and the right mean of the untreated blanks do not differ by more than 15% from the mean of all untreated blanks.
3. If viability is reduced to $< 70\%$ of the blank, it has a cytotoxic potential.
4. The average viability of the negative control group should be judged as non-cytotoxic.
5. The average viability of the positive control group should lower than that of the blank control group, and is judged to be cytotoxic.
6. The 50 % extract of the test sample should have at least the same or a higher viability than the 100 % extract; otherwise the test should be repeated.



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D. Archive of the record and data

All records, data and retained sample were preserved in Master Laboratory Co., Ltd.

Preservation locations

1. Records, data, protocol and report

Rm. A209, 2F., No. 2, Sec. 2, Shengyi Rd., Zhubei City, Hsinchu County 302, Taiwan (R.O.C.)

2. Test article (retained sample)

3F., No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan (R.O.C.)

Preservation Period:

All records and data were preserved for six years. Test articles (retained samples) were retained for six years excluding the samples that sponsors require to be fully returned.



RESULTS

To evaluate the cytotoxicity of test article “RELANO WOUND SPRAY” extract, test article extract was tested towards the cell growth, morphology and viability. After cell exposure to the extract for 24 h, the following items were evaluated:

A. Qualitative determination (Appendix 1, Figure 1)

The morphology of B, NC, PC, and Test article group 1 treated L929 cells stained with NR were observed after 3h under inverted microscope (100X). The morphology of B and NC cells showed long spindle shape with obvious lamellipodia and filopodia instead of lysed, rounded shape, and inhibited growth. However, PC-treated cells showed a nearly complete rounded lysed morphology; cell layers were almost completely destroyed, and growth inhibition was observed. The result of test sample showed the same long spindle shape as B and NC. According to the results of microscopic assay, the percentages of rounded or lysed cells of B, NC, PC, and Test article group 1 were evaluated at 0%, 0%, 100%, and 0% respectively. Therefore, the cytotoxicity of B, NC, PC, and Test article group 1 was graded at 0, 0, 4, and 0 (Table 1).

B. Quantitative determination (Table 2)

The cell viability was evaluated after L929 cells treated with B, NC, PC, and Test article group 1, Test article group 2 for 24h through MTT cell proliferation/viability assay. The absorbance of B, NC, PC, Test article group 1, and group 2 at 570 nm were 1.145 ± 0.009 , 1.086 ± 0.043 , 0.196 ± 0.004 , 0.931 ± 0.035 and 0.936 ± 0.075 respectively; the cell viability represented 100%, 95%, 17%, 81% and 82%; the mortality showed 0%, 5%, 83%, 19% and 18%.



CONCLUSION

According to the results of qualitative and quantitative assays, the results (Table 1 and Table 2) showed “Zero” reactivity. Therefore, none *in vitro* cytotoxicity could be considered in the extract solution of the test article “RELANO WOUND SPRAY”.



REFERENCES

1. Good Laboratory Practice for Nonclinical Laboratory Studies (2019) Food and Drug Administration, R.O.C.
2. Good Laboratory Practice for Nonclinical Laboratory Studies. Title 21 of the U.S. Code of Federal Regulations, Part 58 (2019) United States Food and Drug Administration.
3. Biological evaluation of medical devices-part 5: Tests for *in vitro* cytotoxicity. ISO 10993 (2009).
4. Biological evaluation of medical devices-part 5: Tests for *in vitro* cytotoxicity. EN ISO 10993 (2009).
5. Biological evaluation of medical devices-part 12: Sample preparation and reference materials. ISO 10993 (2021).
6. Biological evaluation of medical devices-part 12: Sample preparation and reference materials. EN ISO 10993 (2021).
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8. Mendes SC, Reis RL, Bovell YP, Cunha AM, Blitterswijk CA, Bruijn JD (2001) Biocompatibility testing of novel starch-based materials with potential application in orthopaedic surgery: a preliminary study. *Biomaterials*. 22, 2057-2064.



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Table 1. Cytotoxicity evaluated using neutral red stain

Test item	Cell lysis (%)			Grade
Blank (B)	0	0	0	0
Negative control (NC)	0	0	0	0
Positive control (PC)	100	100	100	4
Test article group 1	0	0	0	0

Table 2. The results of MTT assay for evaluation of cell viability

Test Item	Absorbance (OD _{570 nm})	Viability (%)	Mortality (%)
Blank (B)	1.145 ± 0.009	100	0
Negative control (NC)	1.086 ± 0.043	95	5
Positive control (PC)	0.196 ± 0.004	17	83
Test article group 1	0.931 ± 0.035	81	19
Test article group 2	0.936 ± 0.075	82	18

Table 3. The OD 570 nm of Untreated Blank

	Absorbance (OD _{570 nm})	Mean of all untreated blank	Difference of blank (%)
Row 2	1.114 ± 0.101	1.135	3.861
Row 11	1.157 ± 0.042		

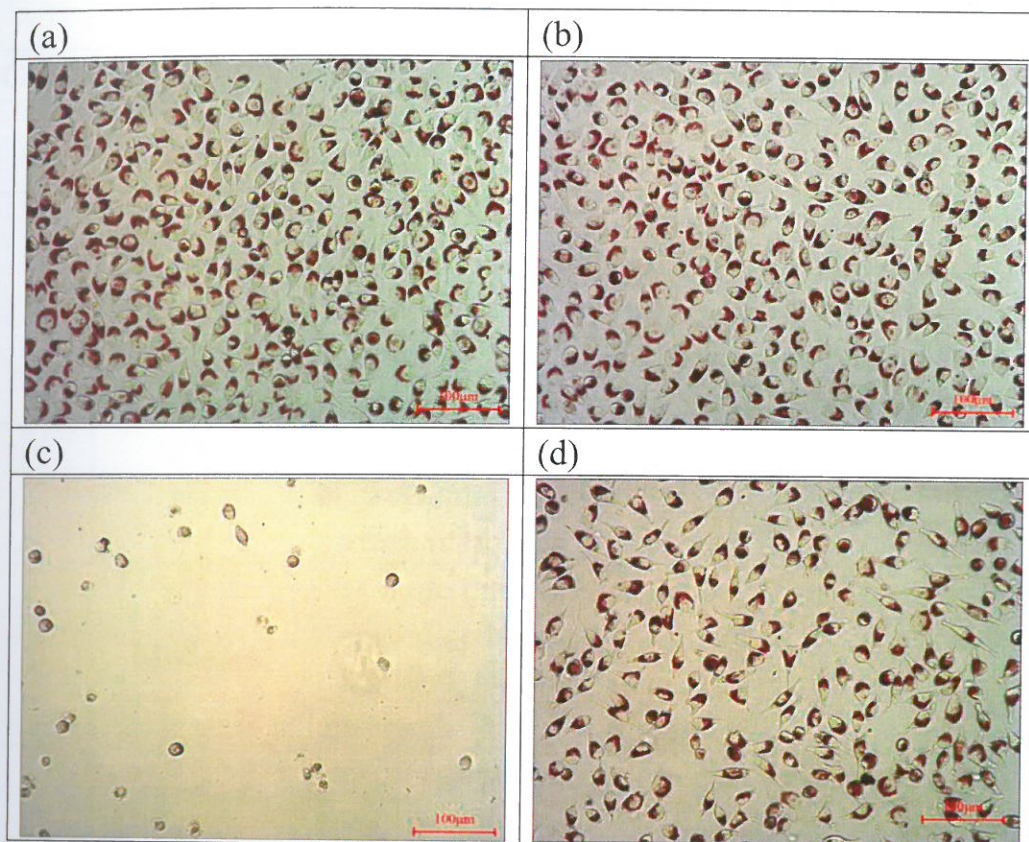


Figure 1. The observed cell morphology of L929 cells after being treated for 24 h under 100X inverted microscope.

- a. Blank (B): culture medium
- b. Negative control (NC): HDPE
- c. Positive control (PC): 10% DMSO
- d. Test article group 1: RELANO WOUND SPRAY

**Appendix 1. Qualitative morphological grading of cytotoxicity**

Grade	Reactivity	Conditions of all cultures
0	None	Discrete intracytoplasmic granules, no cell lysis, no reduction of cell growth
1	Slight	Not more than 20% of the cells are round, loosely attached and without intracytoplasmic granules, or show changes in morphology; occasional lysed cells are present; only slight growth inhibition is observed.
2	Mild	Not more than 50% of the cells are round, devoid of intracytoplasmic granules; not more than 50% growth inhibition is observed.
3	Moderate	Not more than 70% of the cell layers contain rounded cells or are lysed; cell layers not completely destroyed, but more than 50% growth inhibition is observed.
4	Severe	Nearly complete or complete destruction of the cell layers.



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Appendix 2. Test Article Information Sheet

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Information for ☒ Test Article / ☐ Control Article

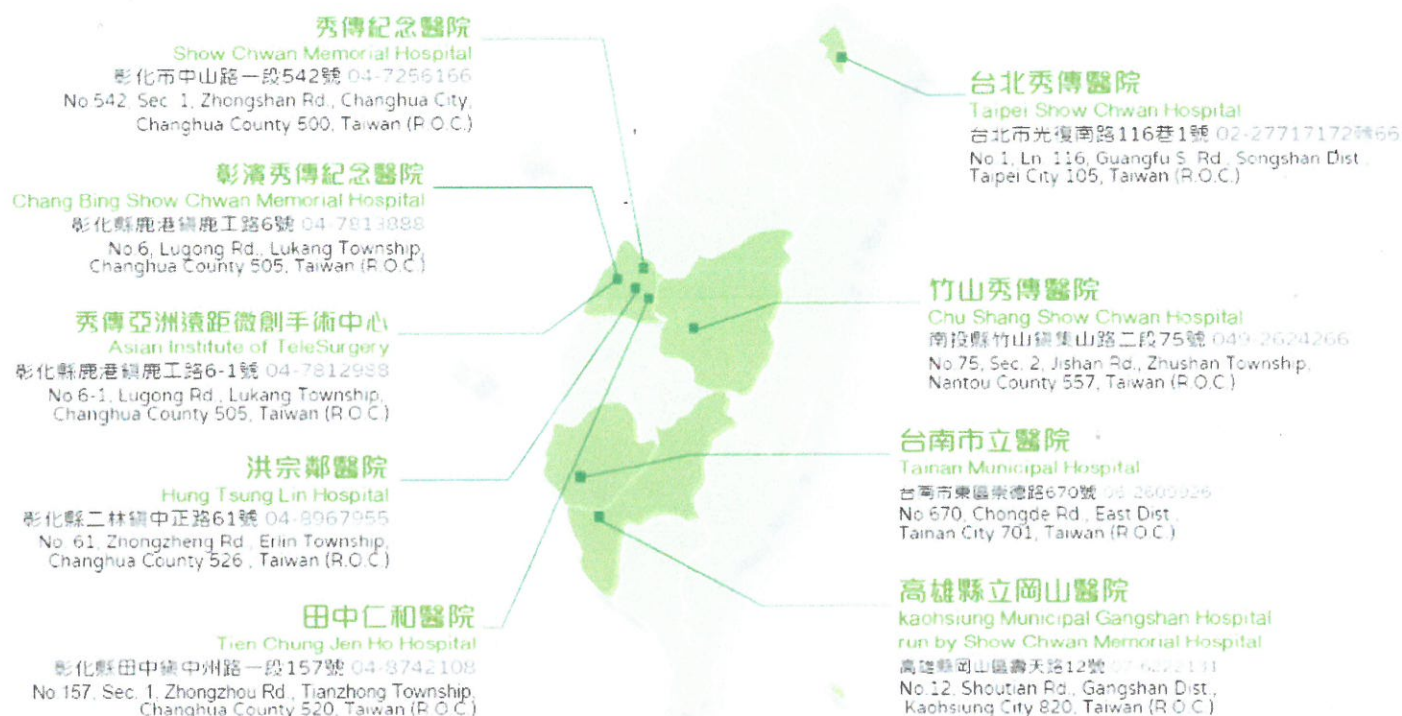
Sponsor Company	Nanoplus Life Biomedical Technology Co., Ltd.
Sponsor Address	9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)
Contract Study item	☒ Base on the contract
Name of test article	RELANO WOUND SPRAY (良諾液態保護噴劑)
Major components	Nano Ion Water, Sodium Carboxymethylcellulose
Sample status	☐ Sterilized (☐ Gamma ☐ EO ☐ Steam) ☒ Not Sterilized
Storage condition	☒ Room temperature (10°C-30°C) ☐ Refrigeration (2°C-8°C) ☐ Others:
Expiry day	05 / 26 / 2025
Specific requirement	None.
Lot number	☒ Base on the specific number on the package: LOT51223050003 ☐ Others:
Extract by	☒ Weight (0.2g/ml) Total weight of each test article: <u>about 50 g (1g/ml)</u> ☐ Surface (Sample thickness: ☐ >1.0mm ☐ 0.5-1.0mm ☐ <0.5mm) Total area surface of each test article:
Absorption	☒ Non absorption ☐ Water absorption rate:/ Oil absorption rate:
Sponsor Signature	



Appendix 3. Test Article Photo







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